

Chapter 1. The Orbits of One-Dimensional Maps.

1.1 Iteration of Functions and Examples of Dynamical Systems.

Dynamical systems is the study of how things change over time. Examples include the growth of populations, the change in the weather, radioactive decay, mixing of liquids and gases such as the ocean currents, motion of the planets, the interest in a bank account. Ideally we would like to study these with continuously varying time (what are called flows), but in this text we will simplify matters by only considering discrete changes in time. For example, we might model a population by measuring it daily. Suppose that x_n is the number of members of a population on day n , where x_0 is the initial population. We look for a function $f : \mathbb{R} \rightarrow \mathbb{R}$, (where $\mathbb{R}$ = set of all real numbers), for which

$$x_1 = f(x_0), x_2 = f(x_1) \text{ and generally } x_n = f(x_{n-1}), n = 1, 2, \dots$$

This leads to the iteration of functions in the following way:

Definition 1.1.1 Given $x_0 \in \mathbb{R}$, the orbit of x_0 under f is the set

$$O(x_0) = x_0, f(x_0), f^2(x_0), \dots,$$

where $f^2(x_0) = f(f(x_0))$, $f^3(x_0) = f(f^2(x_0))$, and continuing indefinitely, so that

$$f^n(x) = f \circ f \circ f \circ \dots \circ f(x); (n - \text{times composition}).$$

Set $x_n = f^n(x_0)$, $x_1 = f(x_0)$, $x_2 = f^2(x_0)$, so that in general

$$x_{n+1} = f^{n+1}(x_0) = f(f^n(x_0)) = f(x_n).$$

More generally, f may be defined on some subinterval I of $\mathbb{R}$, but in order for the iterates of $x \in I$ under f to be defined, we need the range of f to be contained in I , so $f : I \rightarrow I$.

This is what we call the iteration of one-dimensional maps (as opposed to higher dimensional maps $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $n \geq 1$, which will be studied briefly in a later chapter).

Definition 1.1.2 A (one dimensional) dynamical system is a function $f : I \rightarrow I$ where I is some subinterval of $\mathbb{R}$.

Given such a function f , equations of the form $x_{n+1} = f(x_n)$ are examples of difference equations. These arise in the types of examples we mentioned above. For example in biology x_n may represent the number of bacteria in a culture after n hours. There is an obvious correspondence between one-dimensional maps and these difference equations. For example, a difference equation commonly used for calculating square roots:

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)$$

corresponds to the function $f(x) = \frac{1}{2}\left(x + \frac{2}{x}\right)$. If we start by setting $x_0 = 2$ (or in fact any real number), and then find x_1, x_2 etc., we get a sequence which rapidly approaches $\sqrt{2}$ (see page 9 of Sternberg [63]). One of the issues we examine is what exactly is happening here.

Examples of Dynamical Systems 1.1.3

1. **The Trigonometric Functions** Consider the iterations of the trigonometric functions starting with $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \sin(x)$. Select $x_0 \in \mathbb{R}$ at random, e.g., $x_0 = 2$ and set $x_{n+1} = \sin(x_n)$, $n = 0, 1, 2, \dots$. Can you guess what happens to x_n as n increases? One way to investigate this type of dynamical system is to enter 2 into our calculator, then repeatedly press the 2nd, Answer and Sin keys (you will need to do this many times to get a good idea - it may be easier to use Mathematica, or some similar computer algebra system). Do the same, but replace the sine function with the cosine function. How do we explain what appears to be happening in each case? These are questions that we aim to answer quite soon.
2. **Linear Maps** Probably the simplest dynamical system (and least interesting from a chaotic dynamical point of view), for population growth

arises from the iteration of linear maps: maps of the form $f(x) = a \cdot x$. Suppose that x_n = size of a population at time n , with the property

$$x_{n+1} = a \cdot x_n$$

for some constant $a \neq 0$. This is an example of a linear model for the growth of the population.

If the initial population is $x_0 > 0$, then $x_1 = ax_0, x_2 = ax_1 = a^2x_0$, and in general $x_n = a^n x_0$ for $n = 0, 1, 2, \dots$. This is the exact solution (or closed form solution) to the difference equation $x_{n+1} = a \cdot x_n$. Clearly $f(x) = ax$ is the corresponding dynamical system. We can use the solution to determine the long term behavior of the population:

x_n is very well behaved since:

- (a) if $a < 1$, then $x_n \rightarrow 0$ as $n \rightarrow \infty$,
- (b) if $0 < a < 1$ then $x_n \rightarrow 0$ as $n \rightarrow \infty$ (i.e., the population becomes extinct),
- (c) if $a = 1$, then the population remains unchanged.

3. Affine maps These are functions $f : \mathbb{R} \rightarrow \mathbb{R}$ of the form $f(x) = ax + b$ ($a \neq 0$), for constants a and b . Consider the iterates of such maps:

$$f^2(x) = f(ax + b) = a(ax + b) + b = a^2x + ab + b,$$

$$f^3(x) = a^3x + a^2b + ab + b,$$

$$f^4(x) = a^4x + a^3b + a^2b + ab + b,$$

and generally

$$f^n(x) = a^n x + a^{n-1}b + a^{n-2}b + \dots + ab + b.$$

Let $x_0 \in \mathbb{R}$ and set $x_n = f^n(x_0)$, then we have

$$\begin{aligned} x_n &= a^n x_0 + (a^{n-1} + a^{n-2} + \cdots + a + 1)b \\ &= a^n x_0 + b \left(\frac{a^n - 1}{a - 1} \right), \text{ if } a \neq 1 \end{aligned}$$

or

$$x_n = \left(x_0 + \frac{b}{a - 1} \right) a^n + \frac{b}{1 - a}, \text{ if } a \neq 1$$

is the closed form solution in this case (here we have used the formula for the sum of a finite geometric series:

$$\sum_{k=0}^{n-1} r^k = \frac{r^n - 1}{r - 1}$$

when $r \neq 1$). If $a = 1$, the solution is $x_n = x_0 + nb$. We can use these equations to determine the long term behavior of x_n . We see that:

(a) if $|a| < 1$ then $a^n \rightarrow 0$ as $n \rightarrow \infty$, so that

$$\lim_{n \rightarrow \infty} x_n = \frac{b}{1 - a}$$

(b) if $a > 1$, then $\lim_{n \rightarrow \infty} x_n = \infty$ for $b, x_0 > 0$,

(c) if $a = 1$, then $\lim_{n \rightarrow \infty} x_n = \infty$ if $b > 0$. The limit won't exist if $b < 0$.

3. **The Logistic Map** $L_\mu : \mathbb{R} \rightarrow \mathbb{R}, L_\mu(x) = \mu x(1 - x)$ was introduced to model a certain type of population growth (see [41]). Here μ is a real parameter which is fixed. Note that if $0 < \mu \leq 4$ then L_μ is a dynamical system of the interval $[0, 1]$, i.e. $L_\mu : [0, 1] \rightarrow [0, 1]$. For example, when $\mu = 4$, $L_4(x) = 4x(1 - x)$, with $L_4([0, 1]) = [0, 1]$ with graph given below. If $\mu > 4$, then L_μ is no longer a dynamical system of $[0, 1]$ as $L_\mu([0, 1])$ is not a subset of $[0, 1]$.

Recurrence Relations 1.1.4

Many sequences can be defined recursively by specifying the first term (or two), and then stating a general rule which specifies how to obtain the n th term from the $(n-1)$ th term (or other additional terms), and using mathematical induction to see that the sequence is “well defined” for every $n \in \mathbb{Z}^+ = 1, 2, 3, \dots$. For example, $n!$ = n -factorial can be defined in this way by specifying $0! = 1$, and $n! = n \cdot (n-1)!$, for $n \in \mathbb{Z}^+$. The Fibonacci sequence F_n , can be defined by setting

$$F_0 = 1, F_1 = 1 \text{ and } F_{n+1} = F_n + F_{n-1}, \text{ for } n \in \mathbb{Z}^+$$

so that $F_2 = 2, F_3 = 5$ etc.

The orbit of a point $x_0 \in \mathbb{R}$ under a function f is then defined recursively as follows:

we are given the starting value x_0 , and we set

$$x_n = f(x_{n-1}), \text{ for } n \in \mathbb{Z}^+.$$

The principle of mathematical induction then tells that x_n is defined for every $n \in \mathbb{Z}^+$, since it is defined for $n = 0$, and assuming it has been defined for $k = n-1$ then $x_n = f(x_{n-1})$ defines it for $k = n$.

Ideally, given a recursively defined sequence x_n , we would like to have a specific formula for x_n in terms of elementary functions (so called closed form solution), but this is often very difficult, or impossible to achieve. In the case of affine maps and certain logistic maps, there is a closed form solution. One can use the former to study problems of the following type:

Example 1.1.5 An amount \$ T is deposited in your bank account at the end of each month. The interest is r per month (Assume $A(0) = T$).

Answer. $A(n)$ satisfies the difference equation

$$A(n+1) = A(n) + A(n)r + T, \text{ where } A(0) = T,$$

or

$$A(n+1) = A(n)(1+r) + T,$$

so we set $x_0 = T$, $a = 1+r$ and $b = T$ in the formula of Example 1.1.3, then the solution is

$$\begin{aligned} A(n) &= (1+r)^n T + T \left(\frac{(1+r)^n - 1}{1+r-1} \right) \\ &= (1+r)^n T + \frac{T}{r} ((1+r)^n - 1) \end{aligned}$$

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Remark 1.1.6 It is conjectured that closed form solutions for the difference equation arising from the logistic map are only possible when $\mu = 2, 4$ (see Exercises 1.1–3 for the cases where $\mu = 2, 4$, and 7 for the case where $\mu = -2$ and also [67] for a discussion of this conjecture).

Exercises 1.1

- (a) If $L_\mu(x) = (\mu x)$ is the logistic map, calculate L_μ^2 and $L_\mu^3(x)$.
- (b) Use Example 1.1.3 for affine maps to find the solutions to the difference equations:
 - i. $x_{n+1} - \frac{x_n}{3} = 2, x_0 = 2$
 - (c) $x_{n+1} + 3x_n = 4, x_0 = -1$.
- 4. A logistic difference equation is one of the form $x_{n+1} = \mu x_n(1 - x_n)$ for some fixed $\mu \in \mathbb{R}$. Find exact solutions to the logistic equations:
 - (a) $x_{n+1} = 2x_n(1 - x_n)$. Hint: use the substitution $x_n = (1 - y_n)/2$ to transform the equation into a simpler equation that is easily solved.

- (b) $x_{n+1} = 4x_n(1 - x_n)$. Hint: set $x_n = \sin^2(\theta_n)$ and simplify to get an equation that is easily solved.
5. You borrow \$ P at r month. Write down a difference equation for the amount owing A(n) at the end of each month (so $A(0) = P$). Solve the equation to find a closed form for A(n). If $P = 100,000$, $M = 1000$ and $r = 4$, after how long will the loan be paid off?
6. If

$$T_\mu(x) = \begin{cases} \mu x & \text{if } 0 \leq x < 1/2 \\ \mu(1 - x) & \text{if } 1/2 \leq x < 1 \end{cases}$$

show that T_μ is a dynamical system of $[0, 1]$ for $\mu \in (0, 2]$

7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined below. For each of the intervals given, determine whether f can be considered as a dynamical system $f : I \rightarrow I$:

(a) $f(x) = x^3 3x$,

(i) $I = [1, 1]$, (ii) $I = [2, 2]$.

(b) $f(x) = 2x^3 6x$,

(i) $I = [1, 1]$, (ii) $I = [-\sqrt{\frac{7}{2}}, \sqrt{\frac{7}{2}}]$, (iii) $I = [4, 4]$.

- (c) For the following functions, find $f^2(x)$, $f^3(x)$ and a general formula for $f^n(x)$:

$$(i) f(x) = x^2, (ii) f(x) = |x+1|, (iii) f(x) = \begin{cases} 2x & \text{if } 0 \leq x < 1/2 \\ 2x - 1 & \text{if } 1/2 \leq x < 1 \end{cases}$$

- (d) Use mathematical induction to show that if $f(x) = \frac{2}{x+1}$, then

$$f^n(x) = \frac{2^n(x+2) + (-1)^n(2x-2)}{2^n(x+2) - (-1)^n(x-1)}$$

- (e) Show that a closed form solution to the logistic difference equation when $\mu = -2$ is given by